

Superhuman Safe and Agile Racing through Multi-Agent Reinforcement Learning

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Motivation

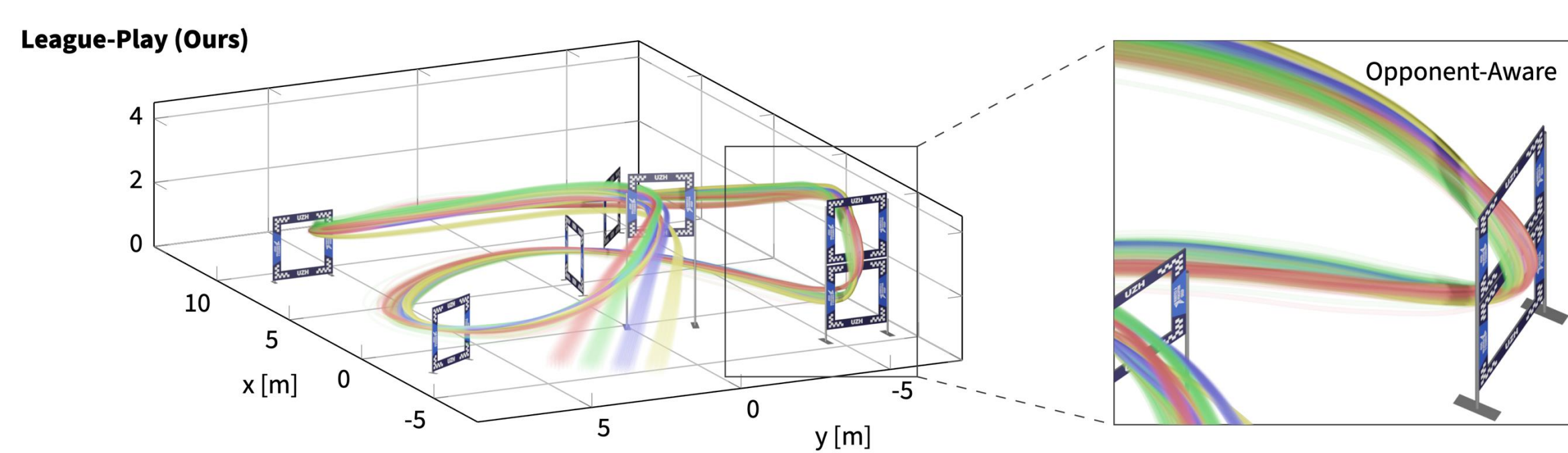
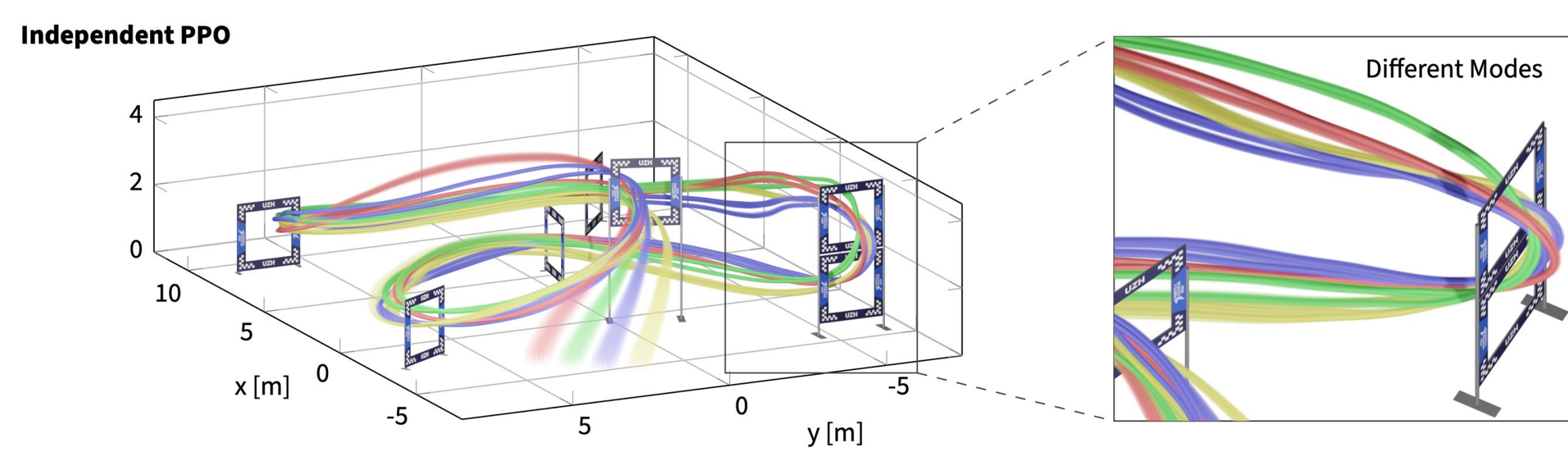
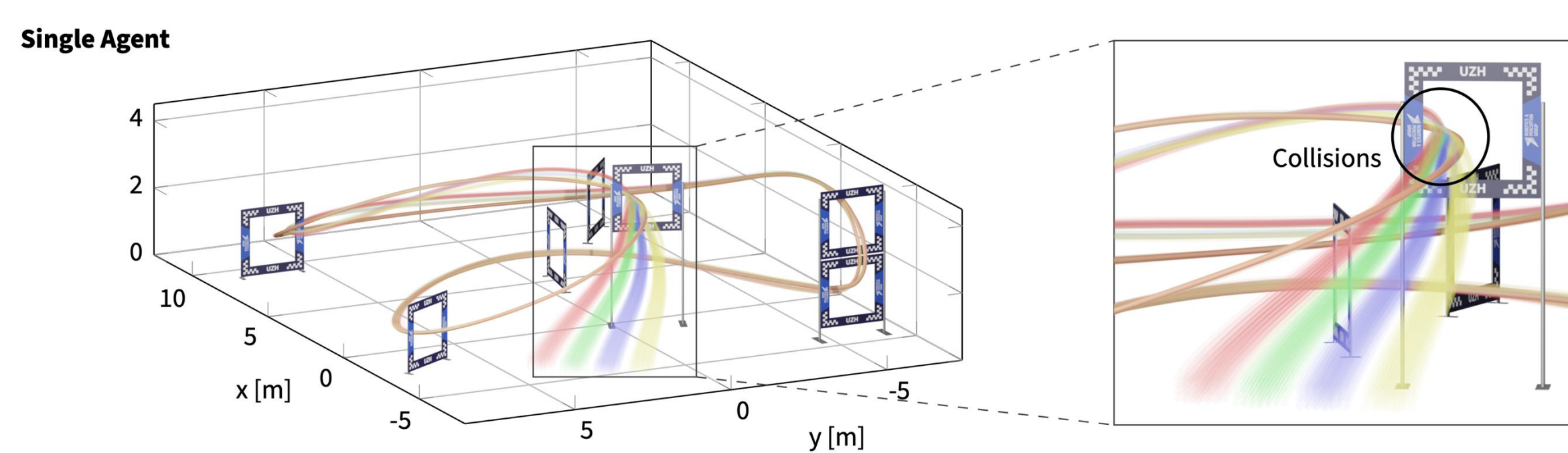
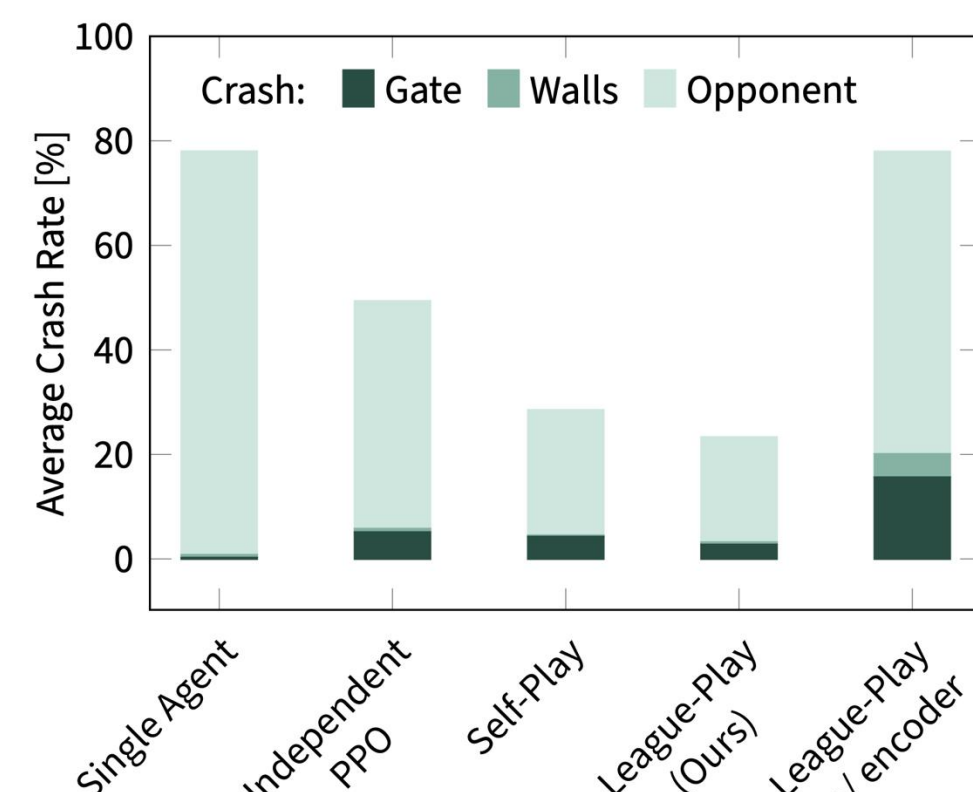
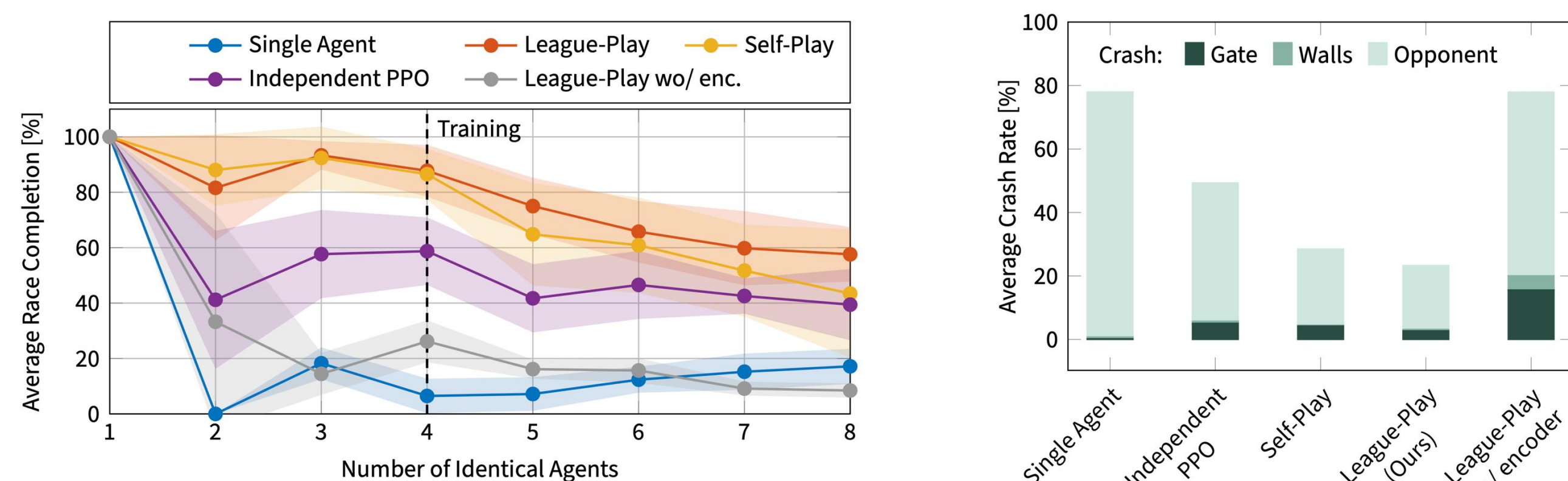
Autonomous systems achieve superhuman performance in isolation, but suffer catastrophic collisions when multiple agents share dynamic environments.

Contributions

We present **superhuman safe multi-player** quadrotor racing through **league-based self-play**. Our agents develop anticipatory collision avoidance and overtaking, outperforming a **champion-level human pilot** at speeds up to **22 m/s** and accelerations of **7 g** while reducing collisions by **over 50%**.

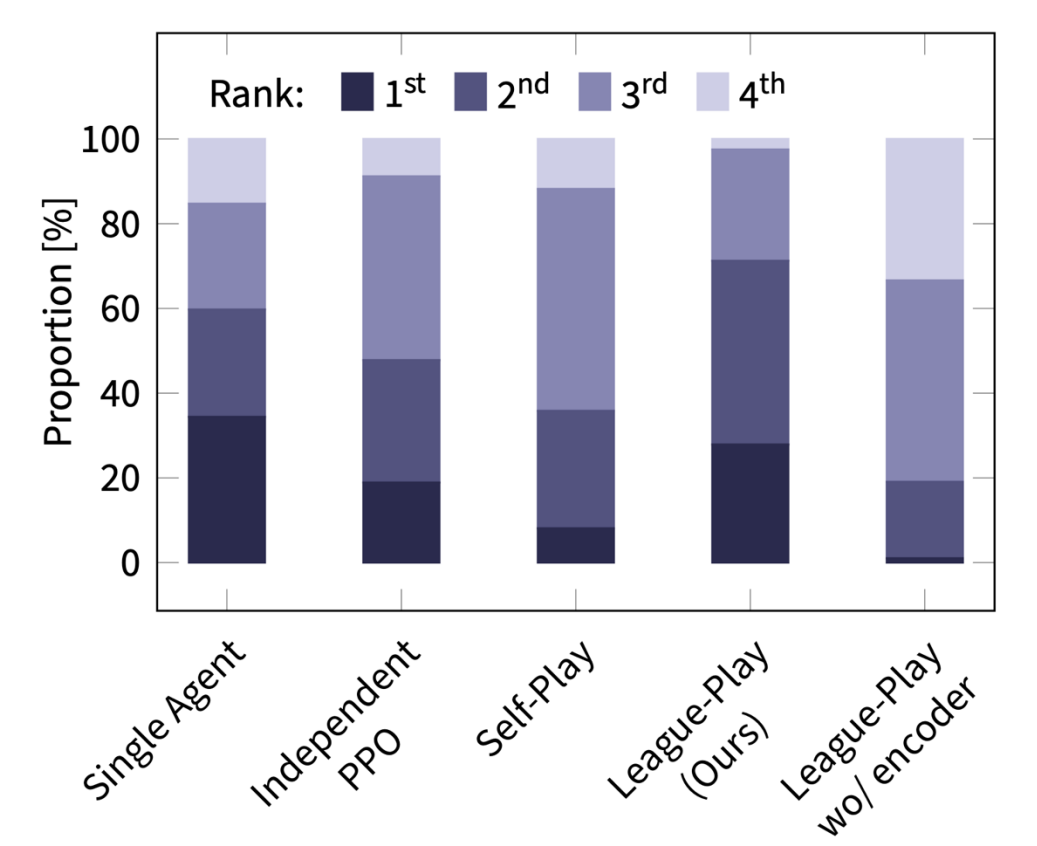
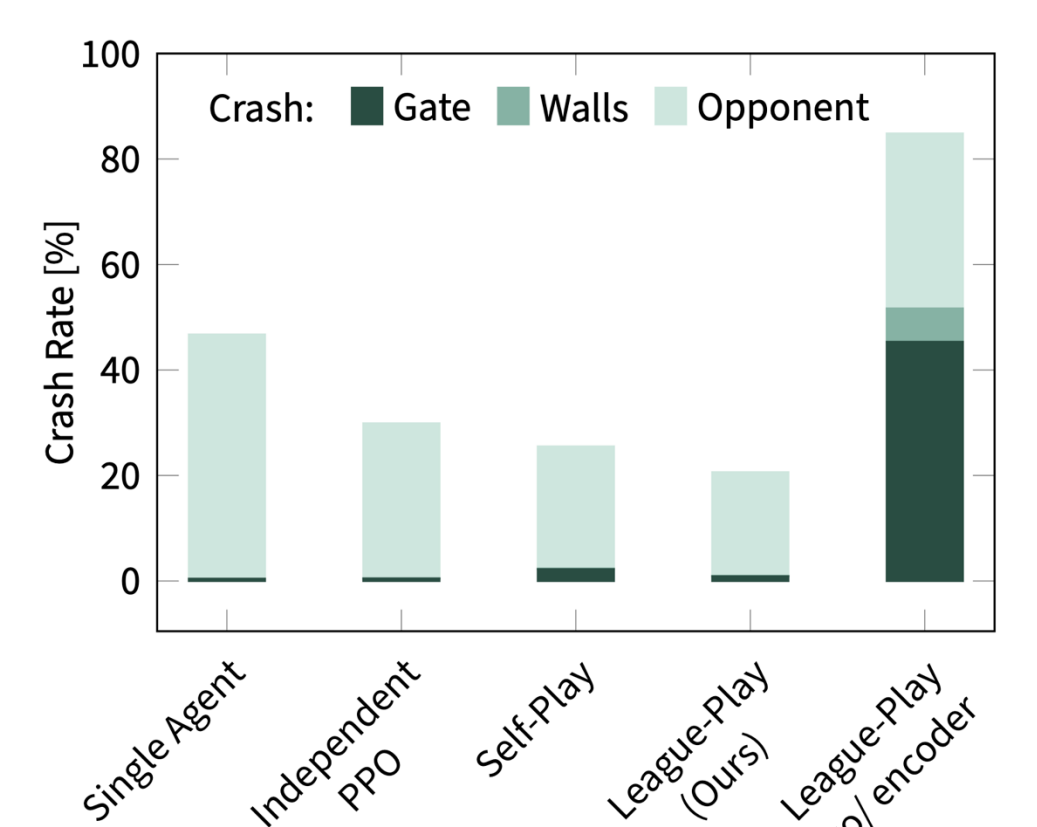
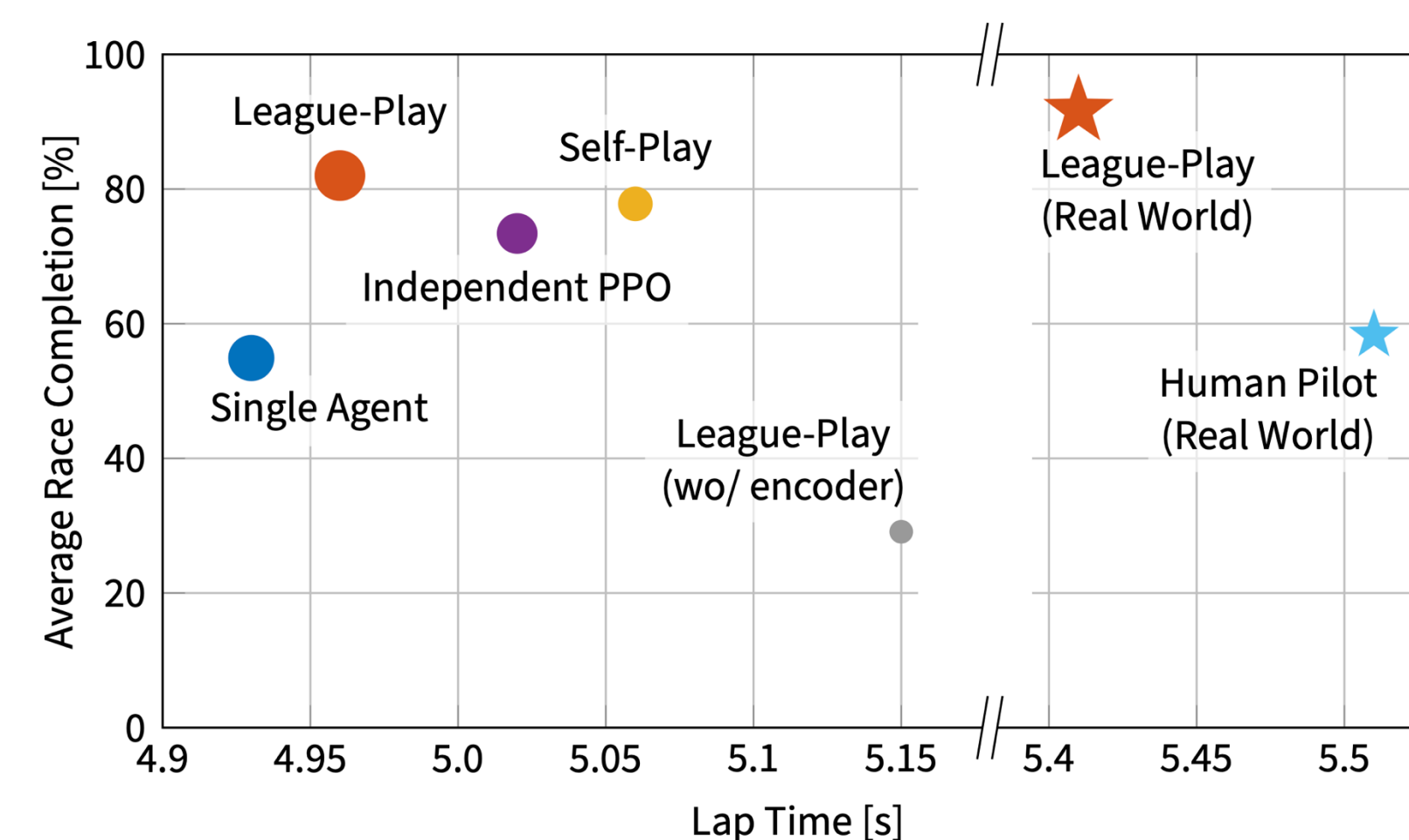
Safe Coordination at Scale

- Single-agent policies select identical racing lines, colliding with **85% probability**
- Independent PPO develops complementary racing modes with co-trained partners, but **fail to generalize** in self-evaluation
- League-play produces **opponent-aware** strategies that remain robust with up to 8 agents, reducing overall crash rates by **over 50%**



Large-Scale Evaluation

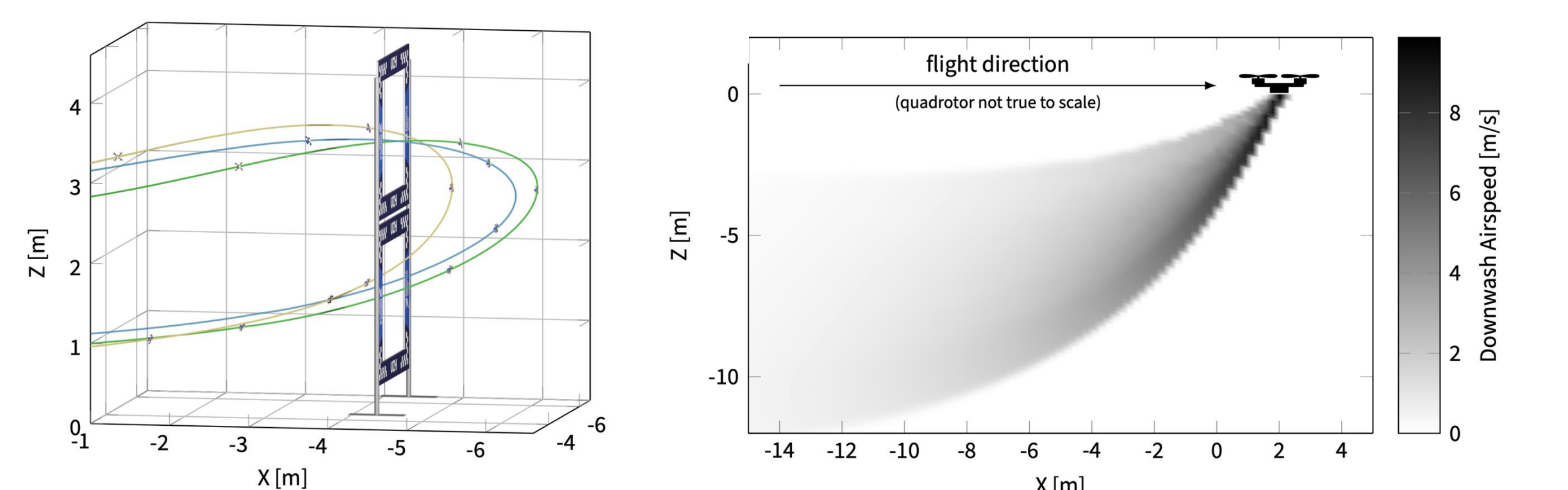
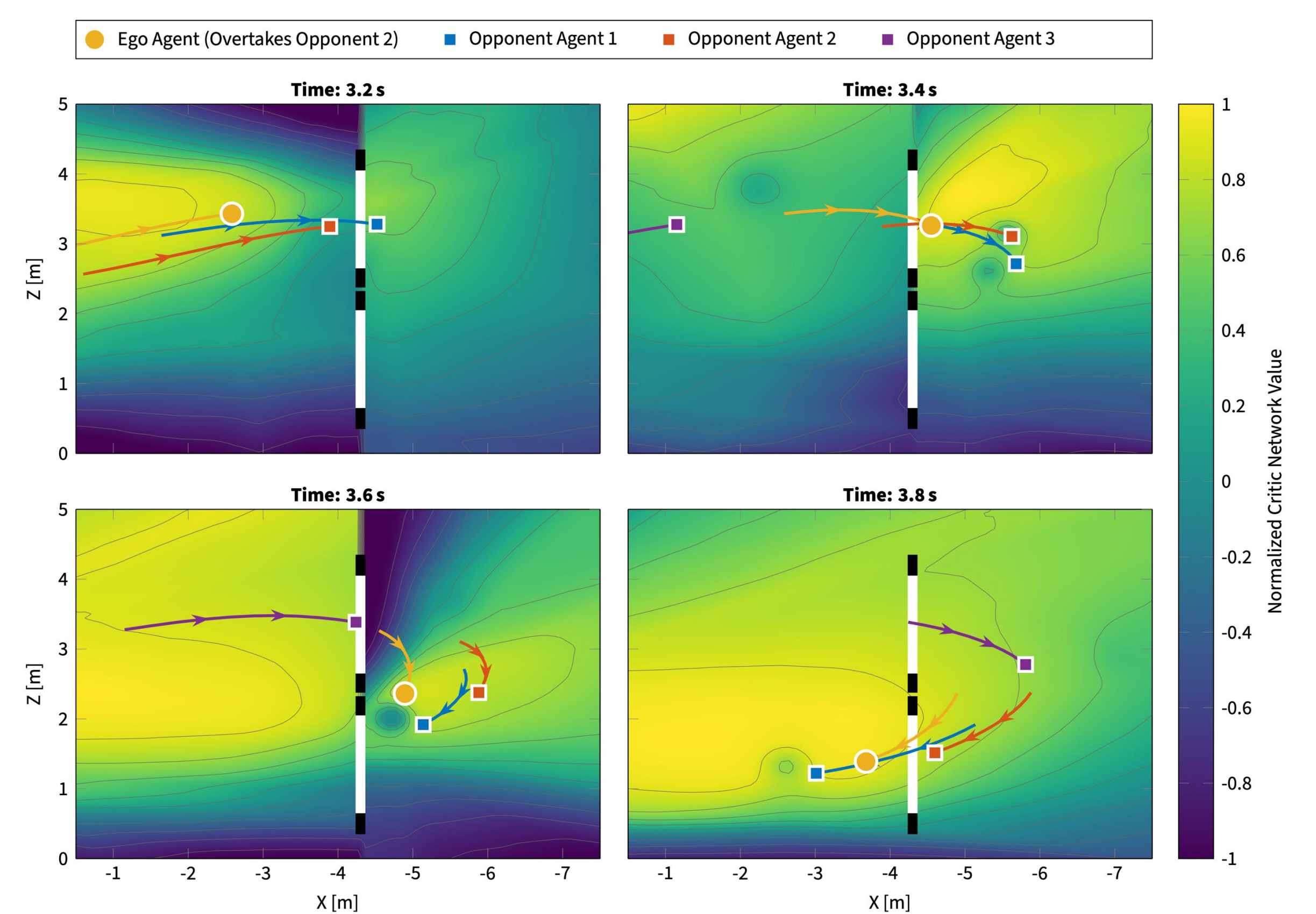
Over **64,000 simulated 4-player races** against diverse opponents. League-play achieves the **highest race completion** while maintaining competitive lap times.



Emergent Anticipatory Behaviors

The learned value function assigns low values **ahead** of opponents along their projected trajectories, revealing anticipatory reasoning rather than purely reactive avoidance.

A **particle-based downwash model** captures aerodynamic interactions during close-proximity flight.



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